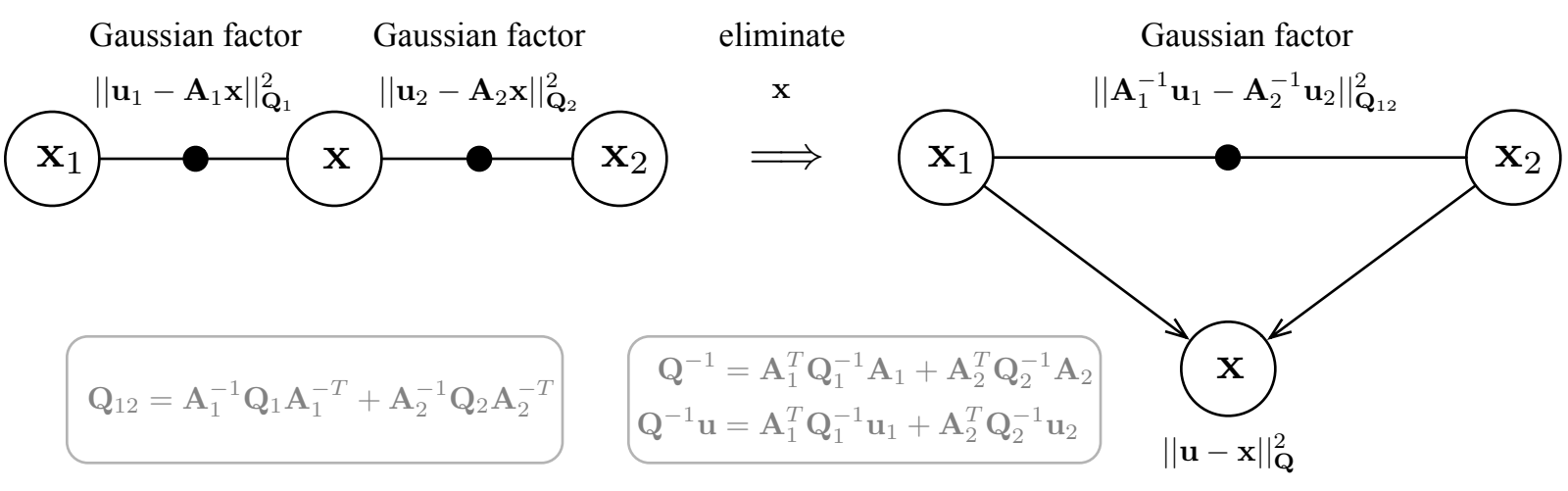

$$\mathbf{Q}_{12} = \mathbf{A}_1^{-1}\mathbf{Q}_1\mathbf{A}_1^{-T} + \mathbf{A}_2^{-1}\mathbf{Q}_2\mathbf{A}_2^{-T}$$

$$\mathbf{Q}^{-1} = \mathbf{A}_1^T\mathbf{Q}_1^{-1}\mathbf{A}_1 + \mathbf{A}_2^T\mathbf{Q}_2^{-1}\mathbf{A}_2$$

$$\mathbf{Q}^{-1}\mathbf{u} = \mathbf{A}_1^T\mathbf{Q}_1^{-1}\mathbf{u}_1 + \mathbf{A}_2^T\mathbf{Q}_2^{-1}\mathbf{u}_2$$